

Hantaviruses Facts over Fear: Risk, Reality, and Misinformation

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SCAN ME



1. INTRODUCTION

Hantaviruses are emerging zoonotic RNA viruses that cause severe diseases such as hemorrhagic fever with renal syndrome (HFRS) and hantavirus cardiopulmonary syndrome (HCPS). Environmental changes, climate change, urbanization and global mobility are expanding their range and increasing outbreak risks. This review provides an evidence-based overview of hantaviruses from virology to public health and future perspectives, separating scientific facts from misinformation.



2. ABSTRACT

Hantaviruses are emerging zoonotic RNA viruses belonging to the family Hantaviridae, responsible for severe human diseases including hemorrhagic fever with renal syndrome (HFRS) and hantavirus cardiopulmonary syndrome (HCPS). These viruses are primarily maintained in rodent reservoirs and transmitted to humans through inhalation of aerosolized viral particles derived from rodent urine, saliva, or feces. Although historically considered geographically restricted pathogens, increasing environmental disruption, climate change, urbanization, and global mobility have contributed to the expansion of hantavirus distribution and the emergence of novel outbreaks. Recent reports involving Andes virus clusters and cruise ship-associated transmission events have renewed global concern regarding the pandemic potential and public health significance of hantaviruses. This review comprehensively summarizes the current scientific understanding of hantaviruses, including viral taxonomy, molecular biology, genome organization, replication cycle, transmission pathways, host reservoirs, pathogenesis, immune responses, clinical manifestations, epidemiology, diagnosis, treatment strategies, vaccine development, viral evolution, ecological determinants, and future perspective. Particular attention is given to the unique human-to-human transmission capability of Andes virus and the implications of ongoing outbreaks in the Americas. The review further discusses advances in molecular surveillance, phylogenetics, antiviral research, and emerging nanobiotechnological approaches that may improve future therapeutic intervention. Given the high mortality associated with severe hantavirus disease and the absence of universally approved antiviral therapies or vaccines, continued global surveillance and interdisciplinary research remain essential.

3. KEY HIGHLIGHTS



Rodent reservoirs are the natural hosts of hantaviruses.



Transmission occurs mainly through inhalation of aerosolized viral particles.



Andes virus can be transmitted from human to human.



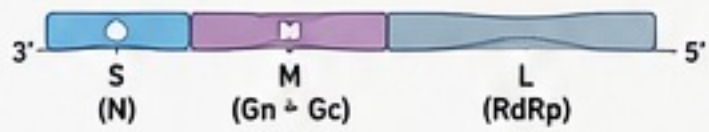
Climate change, land use changes and globalization increase the risk of outbreaks.



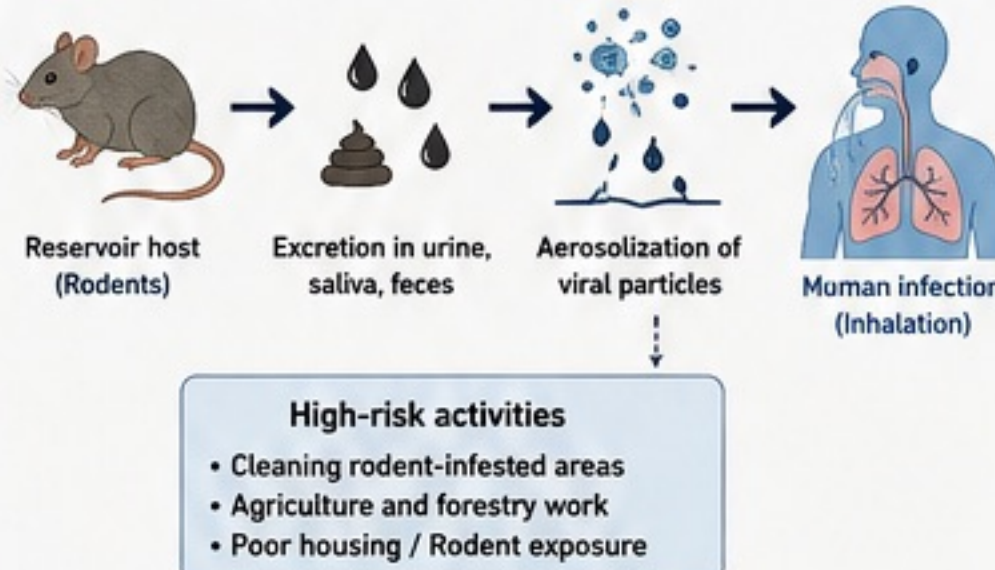
Early diagnosis, surveillance and research are crucial for preparedness and control.

4. VIROLOGY AT A GLANCE

- Family: Hantaviridae
- Genus: Orthohantavirus
- Enveloped, negative-sense single-stranded RNA virus
- Genome: S (N), M (Gn, Gc), L (RdRp)



5. TRANSMISSION CYCLE



6. CLINICAL SYNDROMES

HFRS (Old World)



- Fever
- Hemorrhages
- Renal dysfunction
- Thrombocytopenia

Mortality: 1–15%

HCPS (New World)



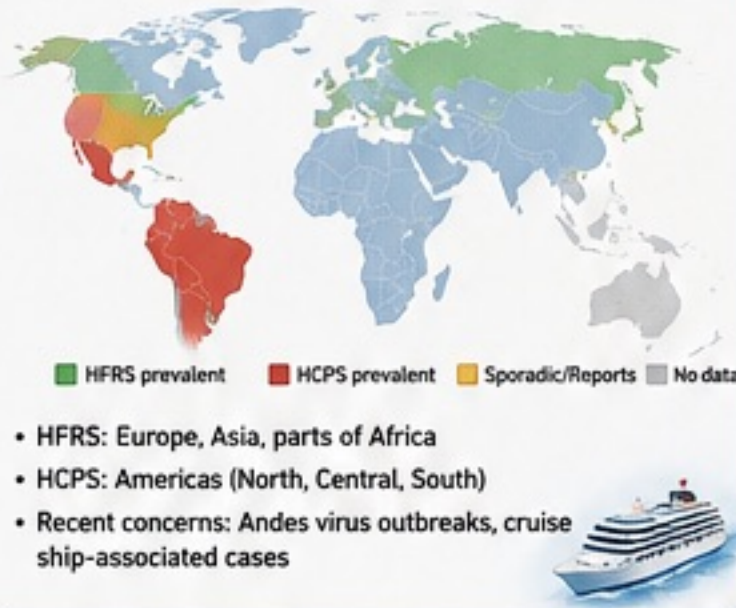
- Fever
- Myalgia
- Respiratory distress
- Pulmonary edema

Mortality: 30–40%

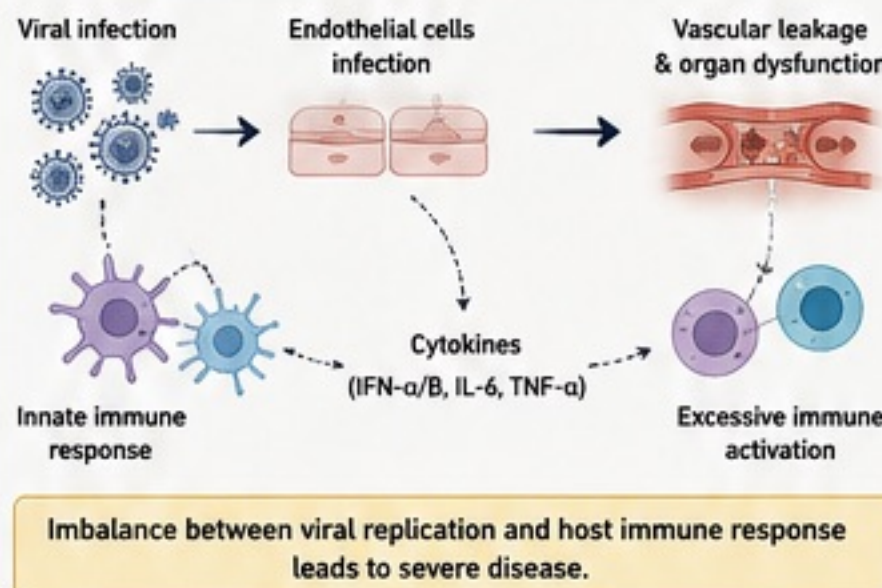


Early supportive care significantly improves outcomes.

7. EPIDEMIOLOGY



8. PATHOGENESIS & IMMUNE RESPONSE



9. DIAGNOSIS & TREATMENT

Diagnosis

- RT-qPCR (early infection)
- IgM/IgG ELISA (serology)
- Immunofluorescence
- Next-generation sequencing



Treatment

- Supportive care
- Fluid management
- Oxygen therapy
- No specific antiviral approved

Research in antivirals and nanobiotechnological approaches is ongoing.

10. PREVENTION & CONTROL

- Rodent control and sanitation
- Use of masks and gloves in high-risk environments
- Proper ventilation of closed spaces
- Hand hygiene and disinfection
- Surveillance and public health preparedness

11. RESEARCH & FUTURE PERSPECTIVES

- Molecular surveillance & phylogenetics
- Nanobiotechnology for diagnostics and therapeutics
- Broad-spectrum antiviral development
- One Health approach for emerging infections

Continuous research and global collaboration are essential to reduce the burden of hantavirus diseases.

12. FACTS OVER FEAR

FACTS

- ✓ Hantaviruses are real and can cause severe disease.
- ✓ Transmission is preventable.
- ✓ Surveillance and research are our strongest tools.
- ✓ Not all rodent contacts lead to infection.

MISINFORMATION

- ✗ Hantaviruses are not new or unknown.
- ✗ They do not spread easily from person to person (except Andes virus).
- ✗ Outbreaks are not inevitable — prevention works.
- ✗ Media exaggeration increases fear, not awareness.

KEYWORDS

Hantavirus; Epidemiology; Transmission; Pathogenesis; Diagnosis; Treatment.

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